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Department of Computer Science and Engineering
PRINCIPLE OF MANAGEMENT AND ECONOMICS (HS251TA)

Synopsis

TITLE: Study how demand forecasting improves product design decisions in software firms		
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A. Abstract

SaaS productivity platforms work in a quickly changing environment where user expectations shift often, and new workflow patterns arise regularly. In this setting, creating the right features at the right time is essential for product success. This study looks at how demand forecasting can be used to improve feature-level product design decisions, focusing on Notion, a popular productivity and workspace platform.

Since internal analytics from SaaS companies are not publicly available, this study develops a forecasting method based on publicly accessible proxy datasets like Google Trends data, app review datasets, and user feedback trends specific to features.

The objective of the research is to explore how forecasting techniques such as exponential smoothing, ARIMA, regression analysis, and sentiment-based trend extraction can predict future interest in features. These insights can then inform real design decisions, including UI/UX changes, feature prioritization, resource allocation, and release planning. The project suggests a structured method that connects demand signals to design actions, showing how forecasting aids data-driven decision-making in today's software companies.

The study aims to create a simple and replicable forecasting framework that software companies can use, even without access to large internal datasets. The overall goal is to demonstrate that effective demand forecasting not only helps companies create better features but also ensures that product design keeps up with user needs, which can reduce churn and improve product adoption.

B. Introduction

Software-as-a-Service (SaaS) products are now crucial for personal productivity, team collaboration, and digital workflows. Among the many tools available, Notion stands out as one of the most flexible platforms. It offers modular databases, templates, workspace organization, and recently, AI-powered features. As the platform grows, product teams face the challenge of deciding which features to design or improve at any time. Relying solely on developer intuition or limited feedback is not enough in today's competitive market. Instead, using data-driven forecasting can provide clear insights into emerging user needs.

Demand forecasting in SaaS means predicting user interest in specific features by examining trends in search behavior, usage patterns, review sentiments, and external events. This is different from forecasting in traditional industries because the demand for SaaS features is nonlinear; it is often swayed by competitor launches, viral content, or sudden shifts in work habits. Recognizing upcoming demand early allows product designers to make better choices regarding UI/UX, workflows, design complexity, and engineering investment.

This study looks at how to adapt forecasting methods to SaaS environments using publicly available datasets. We center on Notion as a case study because of its large user base, active online communities, and open feature discussions. By modeling demand trends for features like Notion AI, templates, and automation workflows, the study shows how forecasting helps guide design decisions. The goal is to demonstrate that forecasting is not just a technical tool; it serves as a strategic foundation for creating products that stay relevant, intuitive, and effective.

C. Literature review

Author	Title	Publication	Summary
Katarzyna Grobler-Dębska, Edyta Kucharska, Bartłomiej Żak, Jerzy Baranowski, and Adam Domagała	Implementation of Demand Forecasting Module of ERP System in Mass Customization Industry—Case Studies	Applied Sciences, November 2, 2022	Proposes a demand forecasting method for the mass customization industry that focuses on predicting demand for key intermediate goods instead of finished products. The study highlights the difficulty in forecasting for highly customizable products (like a complex SaaS tool) and provides an implementation strategy to integrate such forecasting into ERP systems, which is relevant to how product structure (features) influences forecasting.
Meenakshi Agarwal	Streamlining Reports for Demand Planners: A UX Case Study	Medium, January 13, 2025	A User Experience (UX) case study focusing on the design of a robust reporting feature for an AI-based

			demand planning platform. The work emphasizes the importance of providing demand planners with customizable, detailed reports to promote application adoption and guide data-driven design/development decisions, aligning directly with your project's objective of guiding UI/UX design with demand insights.
Alexander-Derek Rein and Jürgen Münch	Feature Prioritization Based on Mock-Purchase: A Mobile Case Study	Lean Enterprise Software and Systems: 4th International Conference, December 2013	Presents an innovative approach to feature prioritization by predicting the real business value of a feature <i>before</i> implementation. The method uses mock-purchases within a mobile app to measure user willingness to pay and extract true business value and an adequate price point, which supports your objective of establishing correlations between demand forecasts and development priorities.
Gursimran Kaur and Manjit Singh Wadhwa	A Hybrid Approach for Feature Extraction From Reviews to Perform Sentiment Analysis	Machine Learning Algorithms and Applications, 2021	Proposes a hybrid approach using Topic Modeling and Natural Language Processing (NLP) tools to automatically extract significant product features from textual reviews and user feedback. This

			methodology aligns directly with your project's plan to use Sentiment & Text Analysis to extract the most-requested features from public review datasets.
Forecastio (Staff)	Time Series Forecasting: Mastering Predictive Sales Models [2025]	Forecastio Blog, September 29, 2025	A comprehensive guide outlining the structured 6-step process for Time Series Forecasting , including data preparation, model selection (such as ARIMA and Exponential Smoothing), and continuous validation. It emphasizes using time-stamped historical data to identify trends and seasonality for strategic business decisions, supporting the Time-Series Forecasting methods listed in your project plan.

D. Objectives

- To understand how interest in different Notion features changes over time by studying publicly available data like Google Trends and user reviews.
- To use forecasting methods such as Exponential Smoothing, ARIMA, and Regression to predict which Notion features are likely to become more or less popular in the future.
- To explore how these demand predictions can guide real product decisions, such as which features should be improved, prioritised, or redesigned.
- To build a practical and easy-to-follow framework that software teams can use to connect demand insights with better UI/UX design and release planning.

E. Hypothesis

This study is based on the idea that forecasted demand patterns significantly influence product design decisions in software firms. It assumes that when a Notion feature shows rising demand signals, such as data from Google Trends, user reviews, and sentiment analysis, product teams are more likely to prioritize, improve, or redesign it. The study also suggests that time-series forecasting models like Exponential Smoothing and ARIMA can reliably predict these demand trends. This allows for data-driven planning. Overall, the main belief is that incorporating demand forecasting into the design cycle leads to decisions that align more closely with user needs, reduces redesign efforts, and increases the adoption of new or improved features in SaaS products.

F. Methodology

The methodology for this study consists of four sequential stages:

Stage 1: Data Collection

Public data representing demand for Notion features is collected from:

- Google Trends (search interest)
- Kaggle review datasets (user sentiments)
- App metadata datasets (ratings, installs)

These datasets act as proxies for internal analytics.

Stage 2: Data Preprocessing

Data is cleaned and organized into feature-specific time series:

- Notion AI search interest (monthly)
- Notion templates interest (monthly)
- Review sentiment scores
- Feature mention counts from reviews

The data is normalized to allow comparison between features.

Stage 3: Forecasting Techniques Applied

Three forecasting methods are used:

- Exponential Smoothing – captures short-term demand shifts
- ARIMA Models – captures long-term patterns and seasonality
- Linear Regression – evaluates external factors influencing demand

Each model generates a predicted demand curve for selected features.

Stage 4: Linking Forecasts to Design Decisions

The outputs are interpreted in terms of design:

- Features with rising interest → prioritize UI/UX optimization
- Features with stable high demand → allocate more engineering effort
- Features with declining demand → reassess their placement in UI

This creates a structured framework connecting forecasting to product design.

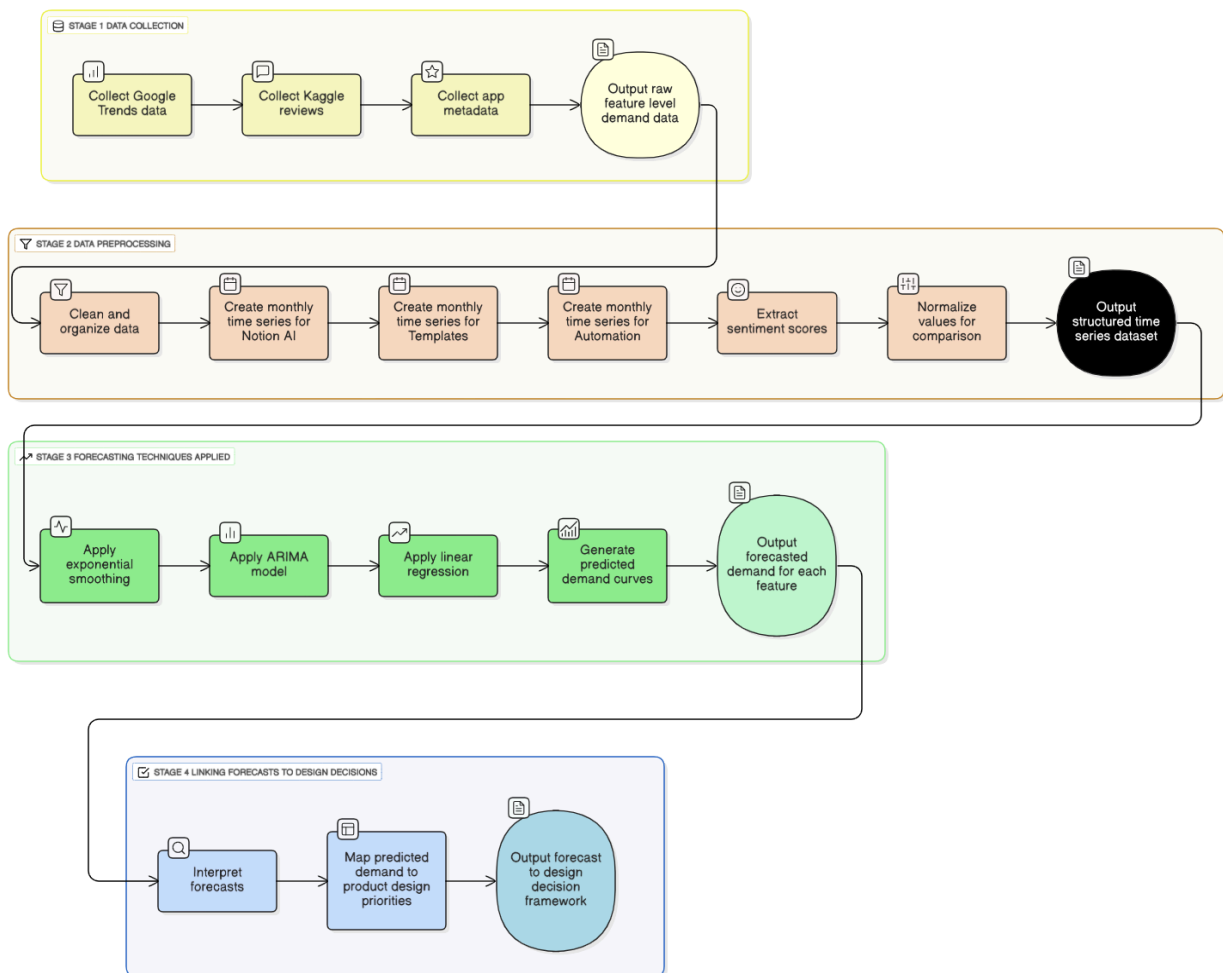


Fig : Methodology flow

G. References:

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